

MINISTRY OF EDUCATION, GUYANA
NATIONAL GRADE SIX ASSESSMENT

SCIENCE
PAPER 02
KEY AND MARK SCHEME
AUGUST 2021

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Question	Possible Responses	Instructions	Marks		
			K	C	A
1 (a)	Meaning of Herbivores: - An animal that feeds on plants	1 mark for correct response	1		
(b)	The organism in the food web that makes its own food: - Grass	1 mark for correct response		1	
(c)	Relationship between the lizard and the grasshopper: - The lizard feeds on the grasshopper - The grasshopper of being eaten by the lizard - The grasshopper is the prey, and the lizard is the predator	1 mark for correct response		1	
(d)	(i) Organism most likely to be affected: - Grasshopper	1 mark for correct response			1
	(ii) Reason grasshopper being mostly affected: - the grasshopper is the only insect in the food web	1 mark for correct response			1
Total			1	2	2
General Objective: G4-4.1.					

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Question		Possible Responses	Instructions	Marks		
				K	C	A
2	(a)	(i) Human body system: – The circulatory system	Correct system named – 1 mark	1		
		(ii) Parts labelled P: – Heart	Correct organ named – 1 mark		1	
	(b)	Completion of statement Blood flowing at <u> Q </u> contains a high percentage of oxygen and a low percentage of carbon dioxide while the blood at <u> R </u> contains a high percentage of carbon dioxide and a low percentage of oxygen.	Both letters correctly inserted – 1 mark		1	
	(c)	Response of the circulatory and respiratory systems during exercise: – During exercise the muscles need more oxygen in order to contract and they produce more carbon dioxide as a waste product. – During exercise, the circulatory and respiratory systems work together to get more <u>oxygen to the working muscles</u> and <u>remove the excess carbon dioxide produced</u> from the body.	Each underlined part, 1 mark each for a maximum of 2 marks			2
Total				1	2	2
General Objective: CC1.1., CC1.2.						

General Objective: CC2.1. – 2.3.

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Question		Possible Responses	Instructions	Marks		
				K	C	A
4	(a)	Type of root shown at X: - Fibrous root	1 mark for correct process	1		
	(b)	(i) Type of root found in guava plant: - tap root/Y	1 mark for correct process		1	
		(ii) Type of root found on rice plant - fibrous root/X	1 mark for correct process		1	

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Question		Possible Responses	Instructions	Marks		
				K	C	A
4	(c)	Difference between the root system at X and at Y: - A tap root generally grows straight down from a plant in order to collect water and minerals from deep in the soil. - Tap roots tend to grow very deep into the ground and often have many small branching tendrils that extend from the main root. - Plants with net vein leaves tend to have tap roots - The fibrous root system occupies less surface area than taproot. - The Taproot system anchors the plant more firmly than the fibrous root. - A tap root is the primary root of the plant while the fibrous root is not. - The fibrous root develops from the stem; this is not the case for the taproot. - The taproot has a few lateral roots that develop from the taproot; this is not the case for the fibrous root. - Taproot is found in dicots, while a fibrous root is found in monocots. - Fibrous root is found in most grasses, while taproot is not found in most grasses. - A taproot can sometimes act as a storage organ for food, while fibrous roots cannot store food.	1 mark for correct response			1
			1 mark for correct response			1

Question		Possible Responses	Instructions	Marks		
				K	C	A
4	(c)	- Fibrous root holds many soil particles together at the surface of the soil; this is not the case with the taproot. - The taproot can reach water deep beneath the surface of the soil; this is not the case for the fibrous root.				
Total				1	2	2
General Objective: CC3.6., CC3.7.						

General Objective: 6.1. – 6.5.

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Question		Possible Responses	Instructions	Marks		
				K	C	A
6	(a)	Forms of energy obtained: - Heat - Light	1 mark for each correct response		2	
	(b)	Source of energy for the candles: - candle wax - fossil fuel	1 mark for correct response	1		
	(c)	Importance of conserving electricity: - To reduce energy costs - To reduce the amount of carbon dioxide is released into the atmosphere. - Reduce global warming/fewer greenhouse gases. - Less use of fossil fuels - Less pollution - Saving habitats for plants and animals - Protect animals and marine wildlife - Safe drinking water				1

General Objective: CC8.10. G4-8.1. – 8.2.